

Proportions & Cross Products

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A proportion sets two ratios equal: $a/b = c/d$. Cross-multiply so $a \times d = b \times c$, then solve.
Example: $3/4 = x/12 \rightarrow 4x = 36 \rightarrow x = 9$.

Round 1 — Retrieve It

1. What does a proportion state about two ratios?

2. Write the general form of a proportion.

3. Write the cross-products rule for $a/b = c/d$.

4. In $3/4 = x/12$, what two products do you set equal?

Round 2 — Sketch It

Write the proportion $3/4 = x/12$ and draw arrows across the equals sign to show the two cross products. Circle the pair you would multiply first.

Proportions & Cross Products — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Solve for x : $\frac{2}{5} = \frac{x}{20}$.

2. Solve for x : $\frac{x}{6} = \frac{15}{18}$.

3. Solve for x : $\frac{7}{x} = \frac{21}{9}$.

Round 4 — Reason from Evidence

A map uses a scale of 2 cm = 50 km. Set up a proportion and use cross products to find how many kilometers 7 cm represents. Show the proportion and the solving steps.

Proportions & Cross Products — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain why cross-multiplying works to solve a proportion. Use one example that is not printed on the cue card.

C

Connect

Connect proportions to unit rates. How could you solve $\frac{3}{4} = \frac{x}{12}$ by first finding a unit rate instead of cross-multiplying?

E

Extend

Create your own proportion with one unknown. Give the numbers and show the work using cross products ($a \times d = b \times c$).

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Proportions & Cross Products

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. It states that two ratios are equal.
2. $a/b = c/d$.
3. $a \times d = b \times c$.
4. $4 \times x$ and 3×12 , so $4x = 36$.

Sketch It — look for

The proportion $3/4 = x/12$ with diagonal arrows showing $4 \times x$ and 3×12 ; a cross-product pair is circled. Cross products give $4x = 36$, so $x = 9$.

Apply It

1. $5x = 40$, so $x = 8$.
2. $18x = 90$, so $x = 5$.
3. $21x = 63$, so $x = 3$.

Reason from Evidence — look for

$2/50 = 7/x \rightarrow 2x = 350 \rightarrow x = 175$ km. (Equivalent setup $2/7 = 50/x$ also gives 175 km.)

ACE — Extend look-for

Many answers possible. Check that the proportion has one unknown and the work correctly uses $a \times d = b \times c$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.